Muhammad Shahroz

Sheikhupura, Pakistan · +92 316 1605532 · muhammadshahroz657@gmail.com
LinkedIn · GitHub · Portfolio

Research interests: mechanics of materials and composites · computational mechanics · scientific machine learning: surrogate models, physics-informed learning, uncertainty quantification · digital twins and data-driven monitoring

EDUCATION

National University of Sciences and Technology (NUST) *Rawalpindi, Pakistan*
Bachelor of Mechanical Engineering Nov 2020 – Jun 2024

- **Thesis:** *Effects of environmental aging on the mechanical properties of synthetic–natural fiber reinforced hybrid composites* (advisor Dr. Zubair Sajid). [Report]
- **Relevant coursework:** Introduction to Finite Element Analysis, Introduction to Machine Learning, Numerical Methods, Probability & Statistics, Mechanics of Materials I–II, Engineering Materials, Manufacturing Processes, Control Engineering, Instrumentation & Measurement.

Massachusetts Institute of Technology, MIT Emerging Talent *Remote*
Certificate in Computer and Data Science Nov 2024 – Dec 2025

- Selected from about 8,700 applicants for the Foundations Track, then for the year-long certificate: MITx 6.0001 and 6.0002 (Python, computational thinking, data science), Universal AI modules, and a collaborative data-science project. [Project]

HONORS & AWARDS

- **MIT Emerging Talent (2024–25):** selected for the Computer and Data Science certificate, about 2% acceptance. [Certificate]
- **Top 6 worldwide, AI for Connectivity Hackathon (lablab.ai, 2025).** [Event]
- **Stanford Code in Place (2024)** [Link] · **Harvard CS50 Puzzle Day (2025):** 9/9 problems solved. [Link]
- **Aspire Leaders Program (Aspire Institute, 2024)** and **UN Millennium Fellowship (2024).**

RESEARCH EXPERIENCE

NUST Materials Laboratory *Undergraduate Researcher*
Advisor: Dr. Zubair Sajid *[Report]* Jun 2023 – May 2024

- Designed and ran a full-factorial durability study of basalt, carbon and basalt–carbon hybrid epoxy laminates (4 laminates $\times$ 3 aging intervals $\times$ 2 ASTM tests); fabricated 72 specimens by hand lay-up and vacuum-bag curing at 60 °C and cut them to ASTM D7264 (flexure) and D2344 (short-beam shear) dimensions.
- Built two Arduino-controlled aging chambers (hygrothermal and marine immersion) with remote data logging when no commercial chamber was available.
- Ran about 1,000 flexural and interlaminar-shear measurements on a universal testing machine. After 20 days of marine exposure, basalt-containing laminates had lost 20 to 30% of flexural strength and 10 to 20% of interlaminar shear strength, while carbon laminates first gained strength at 10 days before declining.

Independent research (extension of the thesis) *Self-directed*
Competing-process model of hygrothermal aging in epoxy laminates 2026

- Wrote a reproducible Python model that couples Fickian and Langmuir (mobile/bound water) moisture transport with a bounded property model of post-cure, plasticization and irreversible damage, to explain the non-monotonic strength histories reported for carbon/epoxy under wet heat.
- Ran a 512-sample Latin-hypercube global sensitivity study and a sparse-data identifiability experiment with multi-start inverse fitting; showed which kinetic parameters gravimetry can recover and why single-factor time–temperature extrapolation fails near the wet glass transition.
- Every parameter carries a provenance label (measured, literature range, or prior); unit tests cover the Fickian limit, bounds, Arrhenius and dry-limit behaviour. [Code and report]

Independent research *Self-directed*
Air-quality forecasting for Lahore from 20 years of atmospheric reanalysis data *[Manuscript]* 2024

- Assembled a 2003–2022 dataset (about 65,000 rows: $PM_{2.5}$, PM_{10}, CO, NO_2, SO_2, temperature); tested stationarity (ADF), decomposed seasonality (STL) and used walk-forward cross-validation.
- Benchmarked ARIMA, SARIMAX, STL baselines and XGBoost using MSE and dynamic time warping; the best model, ARIMA(0,1,0), reached MSE 3.7×10^{-4}, about 30% lower error than the alternatives; manuscript drafted.

SELECTED PROJECTS

Verified finite-element studies, scripted in Python with Gmsh and CalculiX *2026* *[Code] [Site]*

- Seven scripted finite-element studies, each verified against a closed-form solution with executable pass/fail checks: cantilever element convergence (shear locking and hourglassing reproduced), plate-with-hole K_t sweep within 2% of Peterson, modal and buckling analysis within 0.31% of Timoshenko, pin-fin heat transfer with the energy balance closed, L-bracket design optimization (35-point DOE, response surface, 27% lighter optimum), a review of an AI-generated FEA solution (7 of 8 checks fail), and stress-life fatigue with a rainflow counter.
- Wrote the shared library (Gmsh-to-CalculiX element conversion, consistent nodal loads on quadratic faces, result parsers, closed-form references); unit tests and continuous integration re-run the quick studies on every push.

Machine Atlas, an interactive 3D anatomy of eight machines *2026* *[Live] [Code]*

- Every part generated from its engineering dimensions with running kinematics (slider-crank, cam lobes, true gear ratios, Wankel epitrochoid); section, x-ray and exploded views with part-level engineering notes.

Coursework simulation studies *ANSYS, SolidWorks* *[Disc brake] [Cutting tool]*

- Coupled thermal–structural FEA of a disc brake with frictional heating and convection (mesh convergence, material comparison under cyclic loading); tri-axial stress analysis of a cutting tool matching Mohr-circle results within 2.5%; CFD of a model vehicle compared with wind-tunnel drag and lift at 40 m/s.

Connectivity-demand prediction for education and healthcare (AI for Connectivity Hackathon, top 6 worldwide) *[Code] [Event]*

- Machine-learning models and a Flask API predicting connectivity demand from World Bank education statistics and a geocoded database of sub-Saharan public hospitals.

PROFESSIONAL EXPERIENCE

micro1 *Mechanical Engineering Researcher (contract, part-time)*
Expert task design and evaluation for frontier AI models Sep 2025 – Present

- Designed expert-level mechanical engineering tasks with rubric-graded reference solutions for a professional-task benchmark; tasks grounded in ASTM, ISO, ASME and EN standards with measurable acceptance criteria.
- Built tasks that expose model failures on visual engineering inputs (schematics, graphs, process diagrams) and evaluated agent trajectories on CAD workflows.

Mercor *Software & CAD Expert, promoted to Team Lead, Technology & Engineering Evaluation*
Scientific and engineering image evaluation Jan 2026 – Jun 2026

- Led a team of about 100 domain experts delivering evaluation projects on schematics, process diagrams and scientific images; ran reviewer calibration and quality control.

Turing *Research Analyst*
Multimodal AI evaluation Sep 2024 – Oct 2025

- Evaluated multimodal model outputs for reasoning quality and grounding in technical data and engineering drawings.

DTnEC (Digital Transformation and Engineering Consultancy) *Engineering Intern*
Digital twin of pipeline design Jul 2023 – Mar 2024

- ASME-based pipeline design calculations, flow assurance and life-cycle costing; Python and API scripts linking design data to analysis workflows. Earlier (Jun–Jul 2023): design intern, CEME Centre for Research and Innovation, CAD and FEA of drivetrain parts for an electric-vehicle conversion.

TECHNICAL SKILLS

- **Computational:** finite-element analysis (CalculiX, ANSYS Mechanical and Fluent, ABAQUS), Gmsh, CadQuery, SolidWorks; Python (NumPy, SciPy, pandas, scikit-learn, XGBoost, statsmodels), MATLAB; Git, LaTeX.
- **Experimental:** composite fabrication (hand lay-up, vacuum bagging), ASTM D7264 and D2344 testing, universal testing machine, Arduino and IoT instrumentation, environmental aging.
- **Modelling:** moisture diffusion and degradation kinetics, global sensitivity and identifiability analysis, design of experiments and response surfaces, time-series forecasting.

CERTIFICATIONS, AFFILIATIONS & LANGUAGE

- Machine Learning Specialization (Stanford, Coursera) · Mathematics for Machine Learning (Imperial College London) · Optimization Methods for Energy System Studies (Technical University of Denmark, EIT HEI, 2025).
- Student member, ASME and IMechE · Member, NUST Automotive Group.
- **IELTS Academic (Oct 2024):** 7.5 overall (Listening 9.0, Reading 8.0, Writing 7.0, Speaking 6.0).